

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1815

Colfax 100 kV
March 15, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1815. The Interconnection Customer proposes a 121.0 MW solar photovoltaic facility interconnecting at the Colfax 100 kV substation in the Indiana area. The request was received on July 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 496 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,904,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Buchanan Mi - Bonnet Creek 100kV
Contingency: Gaston Rd - Table Mt 100kV Line
Loading: 122.5% (Rating: 137 MVA)
DFAX: 22.7%
Pre-project loading: 102.5%
2. Facility: Buchanan Mi - Northern 100kV
Contingency: Table Mt - Gaston Rd 100kV
Loading: 121.0% (Rating: 138 MVA)
DFAX: 22.7%
Pre-project loading: 101.2%
3. Facility: Ager - Northern 100kV
Contingency: Table Mt - Gaston Rd 100kV line
Loading: 119.0% (Rating: 162 MVA)
DFAX: 22.7%
Pre-project loading: 102.1%
4. Facility: Grandview Ave - Drewry 100kV
Contingency: Table Mt - Gaston Rd 100kV L
Loading: 117.9% (Rating: 207 MVA)
DFAX: 37.9%
Pre-project loading: 95.7%
5. Facility: Freehaven-Lakepark - Twin Branch 100kV
Contingency: Twin Branch Pv - Parkrose 100kV
Loading: 110.8% (Rating: 300 MVA)
DFAX: 32.1%
Pre-project loading: 97.9%

6. Facility: Twin Branch - Kemper County 100kV
Contingency: Twin Branch Pv - Parkrose Substation 100kV
Loading: 110.1% (Rating: 329 MVA)
DFAX: 32.1%
Pre-project loading: 98.3%
7. Facility: Tortoise - Denver Terminal Substation 100kV line
Contingency: Tortoise - Denver Terminal 100kV Line 2
Loading: 106.2% (Rating: 239 MVA)
DFAX: 30.3%
Pre-project loading: 90.8%
8. Facility: Denver Terminal - Tortoise 100kV Cir. 2
Contingency: Tortoise - Denver Terminal 100kV #1
Loading: 106.2% (Rating: 239 MVA)
DFAX: 30.3%
Pre-project loading: 90.8%
9. Facility: Tortoise - Teeland 100kV Circuit 1
Contingency: Teeland - Tortoise 100kV Ckt 2
Loading: 104.7% (Rating: 265 MVA)
DFAX: 30.7%
Pre-project loading: 90.7%
10. Facility: Teeland - Tortoise 100kV
Contingency: Teeland - Tortoise Substation 100kV #1
Loading: 104.7% (Rating: 265 MVA)
DFAX: 30.7%
Pre-project loading: 90.7%
11. Facility: Teeland - Tble Mt Substation 100kV Circuit 1
Contingency: Table Mt Substation - Teeland 100kV Circuit 2
Loading: 103.5% (Rating: 300 MVA)
DFAX: 32.1%
Pre-project loading: 90.5%
12. Facility: Teeland - Table Mt 100kV
Contingency: Teeland Substation - Table Mt 100kV
Loading: 103.5% (Rating: 300 MVA)
DFAX: 32.1%
Pre-project loading: 90.5%
13. Facility: Table Mt - Gaston Rd 100kV
Contingency: Kemper County - Dalreed Substation 100kV
Loading: 101.2% (Rating: 408 MVA)
DFAX: 69.7%
Pre-project loading: 80.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bonnet Crk - Buchanan Mi 100kV	Rebuild line with higher-capacity conductor	\$1,757,000	14% (\$244,000)
Buchanan Mi - Northern 100kV	Rebuild line with higher-capacity conductor	\$9,335,000	14% (\$1,295,000)
Ager - Northern 100kV	Rebuild line with higher-capacity conductor	\$1,942,000	14% (\$270,000)
Grandview Ave - Drewry 100kV	Rebuild line with higher-capacity conductor	\$1,500,000	18% (\$269,000)
Freehaven-Lakepark - Twin Branch 100kV	Rebuild line with higher-capacity conductor	\$1,753,000	9% (\$155,000)
Twin Branch - Kemper County 100kV	Rebuild line with higher-capacity conductor	\$3,597,000	9% (\$318,000)
Tortoise - Denver Terminal 100kV Cir. 1	Reconductor line	\$796,000	9% (\$71,000)
Denver Terminal - Tortoise 100kV Cir. 2	Reconductor line	\$774,000	9% (\$69,000)
Tortoise - Teeland 100kV Circuit 1	Reconductor line	\$508,000	9% (\$45,000)
Teeland - Tortoise 100kV	Reconductor line	\$722,000	9% (\$64,000)
Teeland - Tble Mt Substation 100kV Circuit 1	Reconductor line	\$269,000	9% (\$23,000)
Teeland - Table Mt 100kV L	Reconductor line	\$225,000	9% (\$20,000)
Table Mt - Gaston Rd 100kV	Reconductor line	\$365,000	17% (\$61,000)
	Total allocated to project		\$2,904,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.